NIYOMWUNGERI PARMENIDE ISHIMWE

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with **5+ years** of experience building scalable web applications using React, Node.js, and Python. Currently developing high-performance Deep Packet Inspection (DPI) systems in C and Python on Linux at **Rohde & Schwarz**.

Experienced in system design, code reviews, and software release processes, with active participation in cross-team architecture discussions. Proven ability to deliver end-to-end solutions and optimize performance-critical systems.

TECHNICAL SKILLS

Frontend: React.js, Next.js, Redux, Tailwind CSS, Bootstrap
Backend: Node.js, Express.js, FastAPI, Django
Languages: Python, TypeScript, JavaScript (ES6+), C/C++
Databases: PostgreSQL, MongoDB
Misc: Git, Docker, DevOps, Linux Bash, Agile, Microservices

PROFESSIONAL EXPERIENCE

Software Developer - Rohde & Schwarz

March 2024 – Present

- Develop and maintain high-performance Deep Packet Inspection (DPI) libraries in C and Python on Linux, focusing on efficiency and scalability.

- Implement and optimize Industrial IoT network protocols for real-time traffic analysis.

- Participate in an Architecture Community of Practice, contributing to cross-team discussions on system design, performance, and scalability.

- Conduct code reviews, ensuring code quality, maintainability, and adherence to engineering standards.

- Contribute to software release processes, including testing, validation, and deployment of production-ready features.

- Collaborate in agile teams to deliver secure, high-performance network analysis solutions.

Research Associate - Carnegie Mellon University

October 2023 – February 2024

- Built data pipelines and interactive D3.js visualizations.
- Developed and deployed data-driven web applications on university servers.

Freelance Full-Stack Developer

January 2019 – July 2022

- Designed and developed full-stack applications using MERN and Flask stacks.
- Built REST APIs, authentication systems, and database architectures.
- Delivered multiple production-ready applications for clients.

Web Developer - Inyanja Space

October 2020 – March 2021

- Developed a Learning Management System frontend using React.js and Redux.
- Translated Figma designs into responsive UI components.

KEY PROJECTS

QuizBlog Rwanda

- Built a full-stack educational platform for exam preparation using React, Node.js, Express.js and MongoDB.
- Designed and implemented REST APIs, authentication, and dynamic quiz systems.
- Designed for scalable backend operations and efficient user interactions.

Elysium Group Website

- Integrated frontend and backend services for business operations.
- Built with Node.js, Express.js, React.js and MongoDB.

User Account Management System (Microservices)

- Designed a microservices-based architecture for authentication and user management.
- Implemented secure registration, login, and profile verification workflows.
- Built with Node.js, Next.js and PostgreSQL.

EDUCATION

Master of Science in Information Technology - Carnegie Mellon University (2022 - 2024)

Bachelor of Science in Computer Science - University of Rwanda (2014 – 2018)